

## Applications



1

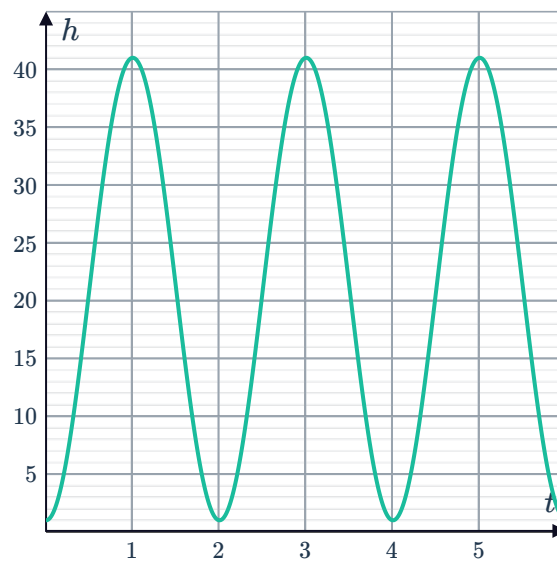
- a Sine, the graph's initial value is at the midline.
- b  $I = \sin(90t)$
- c 210 times

2

- a 6 seconds
- b 2500 mL
- c 200 mL
- d  $60^\circ$  per second
- e  $c = \frac{1}{2}$

3

- a
  - i No
  - ii No
  - iii No
  - iv Yes
- b
  - i 2 minutes
  - ii 1 m
  - iii 41 m
- c

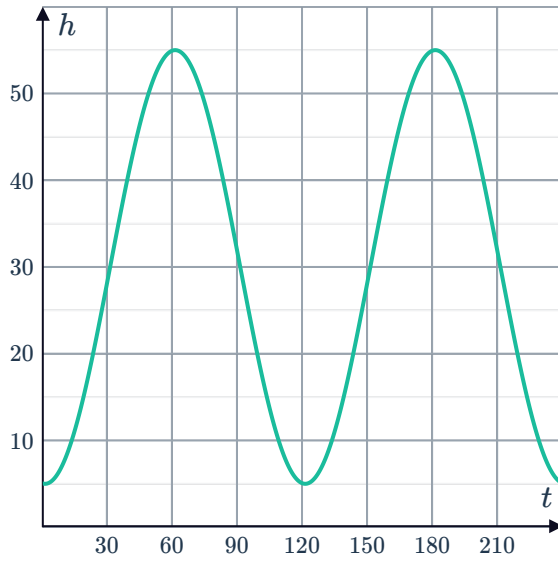


- d  $h = -20 \cos(180t) + 21$
- e  $t = \frac{1}{2}$  minute
- f The height of the passengers above or below the centre of the wheel at time  $t$ .

4



a



b 55 m

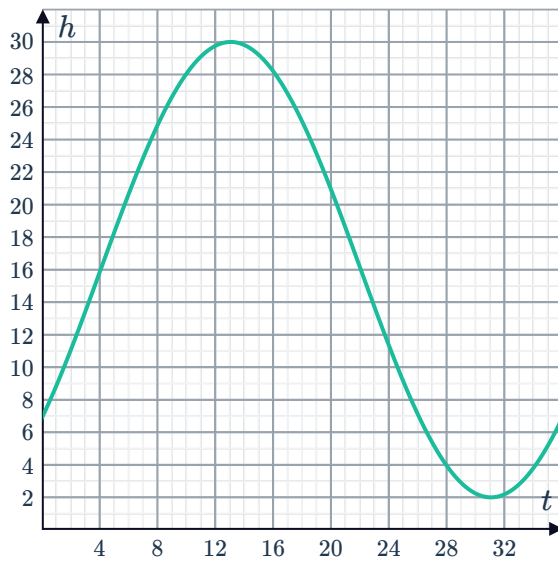
c 5 m

d 36.47 m

e 120 seconds

5

a



b 7.00 m

c  $t = 34.82$  seconds

d 3

e 3.88 m

6

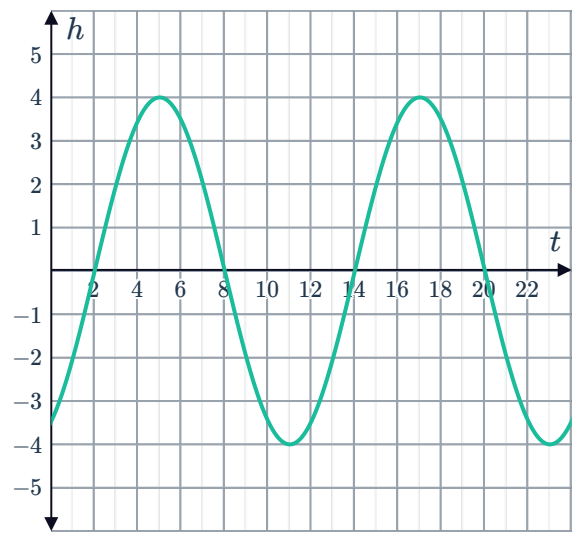
a -2 m

b 5:00 am

c 6 hours



d



e 0 m

f 0.14 m

7

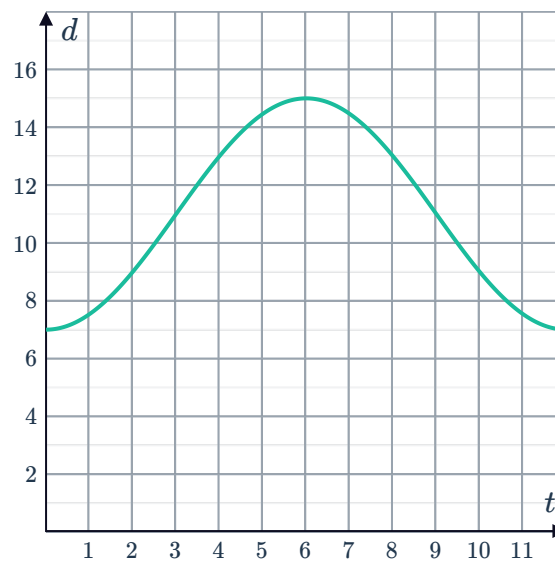
a

| Time ( $t$ , hours)        | 0 | 3  | 6  | 9  | 12 |
|----------------------------|---|----|----|----|----|
| Tide level ( $d$ , metres) | 7 | 11 | 15 | 11 | 7  |

b 4 m

c 12 hours

d



e

i Yes

ii No

iii No

iv No

f  $d = -4 \cos 30t + 11$

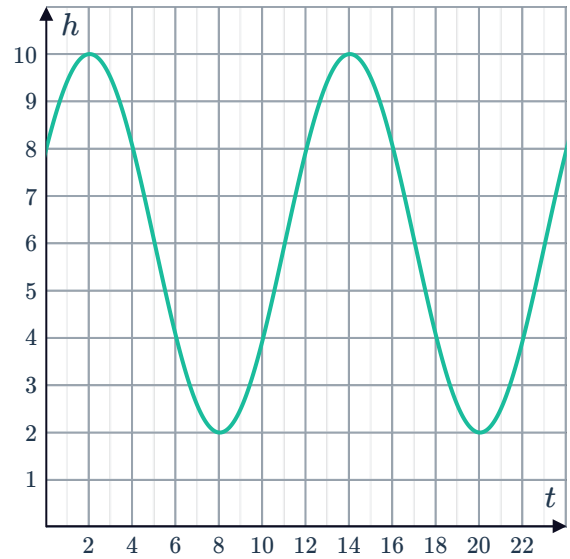
g 3.34 hours

8

a  $h(t) = 6 + 4 \cos(30(t - 2))$



b



c 2:00 am

d 8:00 am

9

a

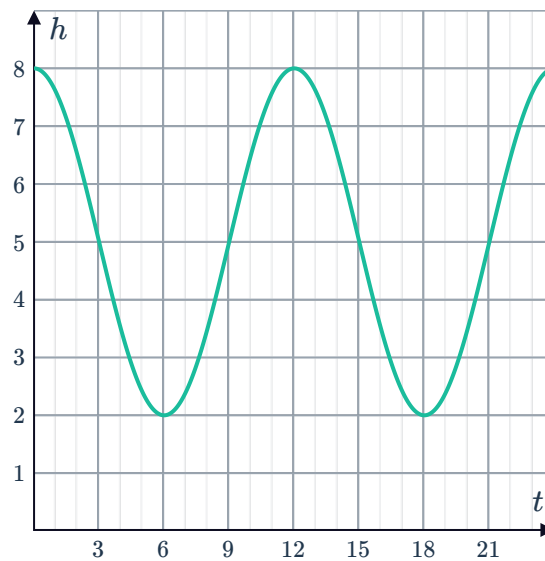
i  $d = 5$  m

ii  $a = 3$  m

iii  $b = 30^\circ$  per hour

iv  $c = 90^\circ$

b



c 6 and 18 hours after noon

10

a

i  $a = 40$  cm

ii  $d = 10$  cm

iii  $c = \frac{1}{4}$

b  $t = \frac{1}{3}$  second

11

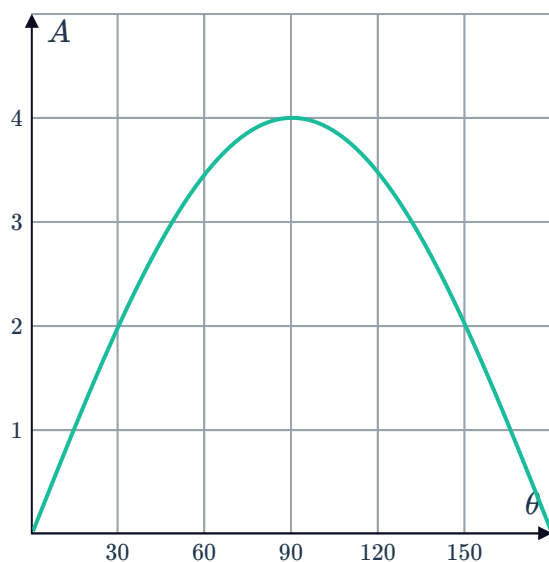
a  $A = 4 \sin \theta$



b  $0 < \theta < 180$

c

d  $30^\circ$



12

a 5 cm to the right of the centre

b To the left

c 1 second

d 4 seconds

e  $x = 5 \cos(90t)$